Dr. Babasaheb Amb[illegible]ar Technological University, Lonere

Summer Examinations May 2015

B.Tech Course in Electronics & Telecommunication Engineering

Subject: Elective I (Electronic Measurements and Instruments)

Semester: IV

Date: 19/05/2015 **Time: 3 Hours** **Max Marks: 70**

Instructions to the Students:

1. **Question No 1 is compulsory and carry 10 marks whereas Question No 2 to Question 7 carries 12 marks each.**
2. **Attempt any five Questions from Question No 2 to Question No 7.**
3. **Illustrate your answers with neat sketches, diagrams etc. wherever necessary.**
4. **Necessary data is given in the respective questions. If such data is not given, it means that the knowledge of that part is a part of examination.**
5. **If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.**

Q.1. Select an appropriate option for each of the following: (10)

1. A pressure gauge calibrated from 0-50kN/m. It has a uniform scale with 100 scale division . One-fifth of a scale division can be read with certainty. The gauge has ....

a) resolution of 0.1 kN/m^2 b)threshold of 0.1kN/m^2 c) dead zone of 0.2kN/m^2 d) resolution of 0.5kN/m^2

2. The sum of all deviations is ....

a) always zero b) never zero c) always positive d) none of these

3. In digital frequency meter, input signal controls enabling and disabling of main gate in.......

a) period measurement b) time interval measurement
c) frequency ratio measurement d) all of these

4. Long persistent screen of CRO is used for .....

a) general purpose applications b)transients c)extremely high speed phenomenon d) all of these

5 The input resistance of cathode ray oscilloscope is of the order of ......

a) tens of ohm b) megaohm c) kiloohm d) fraction of an ohm

Q. 2. Attempt the following (12)

a) Explain accuracy, precision, resolution and linearity of measuring instruments.

b) Explain the thermocouple method of measuring the true RMS value.

Q. 3. Attempt the following (12)

a) What is standard? What are different types of standards?

b) Explain the operation of frequency measurement using digital frequency counter.

Q. 4. Attempt the following (12)

a) Explain vertical amplifier and delay line used in oscilloscope.

b) Explain the working of Digital Storage Oscilloscope with block diagram.

Q. 5. Attempt the following (12)

a) Explain functional block diagram of frequency selective wave analyzer.

b) Draw the block diagram of basic spectrum analyzer using swept receiver design and explain its operation.

Q. 6. Attempt the following (12)

a) Write a note on signal separation device used in network analyzer.

b) Explain the most commonly used receiver techniques in network analyzer.

Q. 7. Attempt the following (12)

a) Explain typical automatic test system with necessary block diagram.

b) Explain computer controlled testing of a radio receiver using block diagram.